Day 115 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Connect Surface Area and Volume Models

Standards: 6.G.A.4, 6.G.A.2

Learning target: I can connect Surface Area and Volume Models, and explain and check my reasoning.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 4, 19

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01 / Learn

A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

A 2 cm by 5 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

1. A closed rectangular prism has two faces of area lw, two of lh, and two of wh.

2. Add the six face areas: SA = 2lw + 2lh + 2wh.

3. Closed surface area = 2(10+6+15) = 62.

4. State the result in the form requested: 62 cm².

Answer: 62 cm²

Worked example 2

A prism measures 11/2 m by 2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 11/2 × 2 × 3/2 = 33/2.

3. The edge counts are 11, 4, 3, giving 132 cubes.

4. Each cube is 1/8 m³

5. 132/8 = 33/2.

Answer: 33/2 m³; 132 cubes

Day 115 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A 8 cm by 6 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

02 / Constructed response

A rectangular prism has edges 4, 3 1/2, and 2 cm. Sketch it and find its volume.

03 / Constructed response

A 7 cm by 6 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A rectangular prism has edges 3, 2, and 5 cm. Sketch it and find its volume.

05 / Constructed response

A 4 cm by 4 cm base and 6 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

06 / Constructed response

A rectangular prism has edges 3 1/2, 1 1/2, and 4 cm. Sketch it and find its volume.

07 / Constructed response

A 5 cm by 3 cm base and 5 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Numeric entry and model reasoning

A prism measures 7/2 m by 2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Two-part numeric entry

A closed box is 2 cm by 2 cm by 5 cm. Part A: enter its surface area. Part B: the 2-by-2 top is removed; enter the remaining surface area and explain.

10 / Constructed response

A rectangular prism has edges 2, 1 1/2, and 1 1/2 cm. Sketch it and find its volume. Show both lwh and Bh.

Day 115 | Unit 08 | Answer key

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1. 180 cm²

Closed surface area = 2(48+24+18) = 180.

6.G.A.4 | Constructed response

2. 28 cm³

B = 4 × 7/2 = 14 cm²; V = B × 2 = 28 cm³.

6.G.A.2 | Constructed response

3. 162 cm²

Closed surface area = 2(42+21+18) = 162.

6.G.A.4 | Constructed response

4. 30 cm³

B = 3 × 2 = 6 cm²; V = B × 5 = 30 cm³.

6.G.A.2 | Constructed response

5. 128 cm²

Closed surface area = 2(16+24+24) = 128.

6.G.A.4 | Constructed response

6. 21 cm³

B = 7/2 × 3/2 = 21/4 cm²; V = B × 4 = 21 cm³.

6.G.A.2 | Constructed response

7. 95 cm²

Closed surface area = 2(15+25+15) = 110. Remove the top area 15: 95.

6.G.A.4 | Constructed response

8. 14 m³; 112 cubes

Multiply edges: 7/2 × 2 × 2 = 14. The edge counts are 7, 4, 4, giving 112 cubes. Each cube is 1/8 m³; 112/8 = 14.

6.G.A.2 | Numeric entry and model reasoning

9. 48 cm²; open box 44 cm²

Closed area = 2(4+10+10) = 48. Subtract exactly one top face, 4, for the open box.

6.G.A.4 | Two-part numeric entry

10. 9/2 cm³

B = 2 × 3/2 = 3 cm²; V = B × 3/2 = 9/2 cm³.

6.G.A.2 | Constructed response